Mengxuan Shen

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Education

Xi'an Jiaotong-Liverpool University **Jiangsu, China**
BEng in Digital Media Technology, GPA: 3.56/4.0 *Aug. 2022–Jun. 2026*
Major Courses: Linear Algebra, Calculus, Java, Electrical Circuits, Electronic Circuits, Engineering Mathematics, Data Structures, Databases, Digital Electronics, Artificial Intelligence, Networking, Continuous and Discrete Time Signals and Systems, Computer Graphics, Analogue and Digital Communications, Advanced OO Programming, Human-Centric Computing, Microprocessor Systems, Mobile Computing, Multimedia Security Systems, Image Processing, Embedded Computer Systems, Computer Auditory Systems.
Certifications: MOS Certificate.

Internship Experience

Vision System Assistant, Hebei Beifang Welding Tools Co., Ltd. *Jul. 2025–Sep. 2025*
- Addressed quality inspection bottleneck where manual checking of robotic welding seams took 3-5 minutes per electrode holder, joined team discussions proposing to develop an image-based pre-screening tool that could flag obvious defects before human verification.
- Collected 450 weld images using workshop industrial camera over three weeks, wrote Python scripts with OpenCV to extract seam regions through edge detection and morphological operations, manually labeled crack, porosity, and incomplete fusion patterns to build ground truth dataset for algorithm development.
- Worked with senior engineer to implement defect detection pipeline combining Canny edge detector for crack identification and contour analysis for porosity detection, iteratively tuned threshold parameters based on feedback from two QC inspectors who reviewed 120 test cases.

Defect Analysis Intern, Hebei Jincheng Yuan Electric Power Tools Co., Ltd. *Jul. 2024–Oct. 2024*
- Tackled systematic false alarm problem (30% rate) in newly deployed automated inspection system for insulated tool handles by extracting and analyzing 200 misclassified images from system database, used histogram analysis to quantify brightness distribution differences between true defects and false positives.
- Implemented preprocessing pipeline using Python and OpenCV to test noise reduction strategies, compared Gaussian blur, median filtering, and bilateral filtering effects on detection stability, found that adaptive histogram equalization combined with morphological closing reduced lighting-induced false alarms by 45% in controlled tests on 150 samples.

Research Experience

Vision-Guided Robotic Control for Hand Rehabilitation *Oct. 2024–Feb. 2025*
Research Assistant, Robotics Laboratory
- Developed real-time visual feedback system for rehabilitation robot that previously operated on fixed trajectories, implementing hand keypoint detection using OpenCV and MediaPipe to track finger joint positions from overhead camera feed.
- Designed position-based visual servoing controller in Python that transformed detected joint coordinates from image space to robot reference frame through calibration matrices, enabling dynamic trajectory adjustment based on actual patient position.
- Integrated visual feedback with robot control loop to enable autonomous path adjustment when subjects shifted posture or relaxed shoulders during exercises, reducing need for manual therapist intervention.

Deep Learning for Medical Image Enhancement *Feb. 2024–Jun. 2024*
Research Assistant, Medical Imaging Group
- Built U-Net based enhancement model to improve diagnostic quality of degraded chest X-rays with poor contrast, motion blur, or inadequate exposure that hindered pneumonia detection.
- Implemented weighted loss function to emphasize reconstruction quality near anatomical boundaries and suspected abnormal regions, addressing limitations of standard approaches that treated all image areas equally.
- Applied data augmentation strategies including elastic deformations and intensity variations to improve model robustness across different anatomical structures and exposure conditions.

Edge Machine Learning for IoT Navigation System *Jan. 2024–May 2024*
Research Assistant, Embedded Systems Lab
- Developed on-device decision tree model for intelligent umbrella system to provide route recommendations based on weather patterns without cloud dependency.

- Collected training dataset using GPS and environmental sensors, recording weather conditions and corresponding navigation patterns to build prediction model deployable on Arduino hardware.

Retrieval-Augmented Generation System for Research Question Answering *May 2023–Sep. 2023*

Research Assistant under Prof. Philipp Koehn (Johns Hopkins University)

- Developed retrieval-augmented generation pipeline to address performance bottleneck in question-answering system where sequential similarity computation across research abstracts resulted in slow query response times.
- Implemented hybrid retrieval architecture combining BM25 for initial filtering followed by Sentence-BERT neural embeddings with FAISS vector database for semantic re-ranking.
- Improved retrieval pipeline efficiency to achieve faster response times while maintaining reasonable accuracy in returning relevant papers within top results.

Project Experience

Intelligent Umbrella for Rainy Day Navigation *Mar. 2025–Jun. 2025*

Team Leader, Jiangsu Provincial College Robot Creativity Competition

- Led a team of four to design and prototype a multifunctional smart umbrella integrating LED lighting, NFC communication, GPS tracking, and 4G connectivity to address rainy-day mobility challenges.
- Implemented core functionalities using Arduino Uno microcontroller, writing C/C++ code for NFC data exchange, GPS location acquisition, and 4G module communication, ensuring stable inter-module coordination.
- Designed umbrella structural components using CAD software and fabricated prototypes via 3D printing, iteratively testing and refining mechanical integration to balance functionality with ergonomic usability.
- Coordinated project workflow from ideation to deployment, managing task allocation, technical troubleshooting, and presentation preparation, successfully demonstrating a working prototype at the competition.

AI Personal Assistant Mobile Application *Mar. 2025–May 2025*

UI/UX Designer

- Designed complete user interface for an AI-powered personal assistant app featuring daily schedule management, personalized diet recommendations, and intelligent suggestion modules.
- Created wireframes, interactive prototypes, and high-fidelity mockups using Figma, defining information hierarchy, icon systems, and transition animations to ensure intuitive navigation flows.

Mathematical Modeling Competition Project *Dec. 2023–Feb. 2024*

Team Leader, Mathematical Contest in Modeling (MCM)

- Led a three-member team in solving interdisciplinary optimization problems, coordinating end-to-end workflow from problem decomposition to solution delivery under 96-hour time constraints.
- Developed data processing and algorithm implementation using Python and Java, performed numerical computations and model validation with MATLAB, fine-tuning parameters to improve prediction accuracy.

Extracurricular Activities

- **All-Subject Tutor**, Private Tutoring Institution *Sep. 2022–Jun. 2025*
- **Head of Publicity**, International Innovation Society *Feb. 2024–Nov. 2024*
- **Translator Volunteer**, International Wushu Invitational Tournament *Mar. 2024–Apr. 2024*
- **Director of Administration & Finance**, Student Union *Feb. 2023–Jun. 2024*
- **Class Monitor**, Xi'an Jiaotong-Liverpool University *Sep. 2022–Jun. 2023*

Awards

- **Second Prize**, Jiangsu Provincial College Intelligent Robot Creativity Competition (Jun. 2025)
- **Honorable Mention (H Award)**, Mathematical Contest in Modeling (MCM) (Feb. 2024)
- **First Place**, ELC Sustainable Development Video Creation Competition (Apr. 2023)
- **Provincial Outstanding Class**, Xi'an Jiaotong-Liverpool University (Jun. 2023)

Professional Skills

- **Programming**: Proficient in **Java**, **Python**; Familiar with **C/C++**
- **AI & Computer Vision**: **PyTorch**, **TensorFlow**, **OpenCV**; CNNs, RNNs, Transformer architectures
- **Design & Multimedia**: **Photoshop**, **Figma**, Premiere, Unity, Canva
- **Tools**: Git, Arduino, MySQL, MATLAB, Excel, Matplotlib